Journal of College Reading and Learning

ISSN: 1079-0195 (Print) 2332-7413 (Online) Journal homepage: www.tandfonline.com/journals/ucrl20

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To cite this article: Élodie Audet, Shelby Levine, Pascale Dubois, Sophie Koestner & Richard Koestner (2023) The Unanticipated Virtual Year: How the Big 5 Personality Traits of Openness to Experience and Conscientiousness Impacted Engagement in Online Classes during the COVID-19 Crisis, Journal of College Reading and Learning, 53:4, 298-315, DOI: 10.1080/10790195.2023.2250402

To link to this article: https://doi.org/10.1080/10790195.2023.2250402

Published online: 07 Sep 2023.

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JOURNAL OF COLLEGE READING AND LEARNING
2023, VOL. 53, NO. 4, 298–315
https://doi.org/10.1080/10790195.2023.2250402

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The Unanticipated Virtual Year: How the Big 5 Personality Traits of Openness to Experience and Conscientiousness Impacted Engagement in Online Classes during the COVID-19 Crisis

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ABSTRACT
The COVID-19 crisis dramatically impacted how academic classes were taught. The present two studies used two-wave prospective longitudinal designs (following two separate cohorts of university students) to examine the predictive role of the Big 5 personality traits of openness to experience and conscientiousness on students' engagement in online classes. Students were asked to report on their levels of motivation and self-efficacy for engagement in online classes. Results suggest that during the Fall 2020 semester the trait of openness to experience may have allowed students to be more engaged in online classes. However, openness to experience was no longer associated with greater engagement during the Winter 2021 semester. Instead, during this second online semester, conscientiousness emerged as the best predictor of heightened engagement in online classes. Interestingly, results suggest that openness to experience and conscientiousness may have different pathways: the benefit of openness to experience was mediated by intrinsic motivation whereas that of conscientiousness by self-efficacy.

KEYWORDS
Conscientiousness; engagement in online classes; intrinsic motivation; openness to experience; self-efficacy

The academic year of 2020–2021 presented a challenge most university students had never encountered before. Given the rapid spread of the COVID-19 virus, many North American universities decided to administer the academic year fully, or at least mostly, online. This was primarily conducted via online work-sharing platforms (e.g., ZOOM), which allowed students to complete classes and assessments from their own homes. The abrupt transition to online classes had started at the beginning of the pandemic in March 2020, yet most university students and teachers perceived it as a temporary situation requiring urgent adjustments based on no planning and relied on teaching strategies like those used in face-to-face classes (Gillis & Krull, 2020). In contrast, both students and teachers had time to see the 2020–

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2021 online academic year coming and prepared accordingly. Thus, the resumption of the online semester in Fall 2020 may have served as an initial adjustment period. However, in the Winter semester, students may have become more accustomed to online classes. This makes the online academic year of 2020–2021 an important time for examining initial versus long-term adjustment to online classes in the context of a global crisis.

The current research examines the predictive role of two of the Big 5 personality traits (openness to experience and conscientiousness) on university students' engagement in online classes and the motivational processes (intrinsic motivation and self-efficacy) that may be implicated. Indeed, the way students reacted and adjusted to online classes, both initially and over time, could partially depend on individual differences in personality traits.

The Big 5 Personality Traits

The Big 5 personality framework suggests that most individual differences in personality can be classified into five broad, empirically-developed domains that are independent of one another (John & Srivastava, 1999). The five personality traits are openness to experience, conscientiousness, extraversion, agreeableness, and neuroticism. The present study focuses exclusively on the Big 5 personality traits of openness to experience and conscientiousness because of their potential relevance to the sudden and novel online academic setting (Abe, 2020). Indeed, the importance of these two personality traits was evidenced in a classic meta-analysis suggesting that individual differences in openness to experience and conscientiousness may influence individuals' adjustment to new environments, both initially and over time (Barrick & Mount, 1991).

Openness to experience is characterized by intellectual curiosity, the tendency to seek novel experiences, and the ability to adjust to change (Schmutte & Ryff, 1997). Scoring high on openness to experience is related to the desire to explore new things (McAdams, 2015). Individuals higher in openness to experience can better adjust to changing situations (Lepine et al., 2000), and this personality trait may be particularly solicited at the beginning of the academic year (Rivers, 2021). In comparison, conscientiousness is defined as the tendency to be organized, reliable, and responsible (McCrae & John, 1992). These were particularly important skills for online learning, and previous studies indicate that conscientiousness is especially important for school performance (Conrad & Patry, 2012; Vianello et al., 2010; Wang et al., 2019).

As mentioned, the importance of these two personality traits was evidenced in a classic meta-analysis of work outcomes by Barrick and Mount (1991), which investigated the relation of the Big 5 personality traits to training and work outcomes. Openness to experience predicted the training (but not work) proficiency (Barrick & Mount, 1991). This suggests that individuals higher in

openness to experience have a tendency toward new experiences in general and are more willing to engage in new (but not familiar) environments. Conscientiousness, however, showed consistent relations with training and work proficiency in every occupation that was assessed (e.g., professionals, police, managers, sales, etc.), depicting this personality trait as something important for the accomplishments of work-related tasks. Indeed, individuals who show a strong sense of persistence, purpose, and obligation generally perform better than those who do not. Similar results have been found in educational environments (Kertechian, 2018; Saklofske et al., 2012).

The results of the Barrick and Mount (1991) meta-analysis indicated that the relation between the Big 5 personality traits and training, as well as later work performance outcomes, could also be seen during the online academic year. Given its novelty, the Fall 2020 semester perhaps served as a "training period," during which students got used to their new learning environment and developed new ways of operating. In contrast, the Winter 2021 semester may have served as a "work period" where students adapted to the online environment, implement these adaptations, and applied their newly formed habits. This suggests that while openness to experience is useful for adjusting to new environments, it may lose its effect on efficiency when the adjustment period is over. In sum, individual differences in openness to experience and conscientiousness may influence how students engage in online classes, both initially and over time.

Motivation, Self-Efficacy, and Engagement in Online Classes

Self-Determination Theory (SDT; Ryan & Deci, 2017) is a useful framework to understand the motivational processes by which personality traits may influence engagement in online classes because it posits there are contextual contingencies that affect motivation and behaviors. SDT researchers highlight the quality as opposed to the quantity of motivation that individuals have. Central to SDT is the distinction between two motivational orientations that impact behavior, wherein the motivation for any behavior is seen on a continuum, ranging from more external (i.e., doing an activity out of feelings of obligation or pressure) to more internal (i.e., a personally meaningful and volitional) reasons (Ryan et al., 2021). *Intrinsic* motivation is seen as the most volitional and autonomous form of all motivations. It involves enacting a behavior for the pure interest, enjoyment, and inherent satisfaction of the activity itself (Holding & Koestner, 2023). When intrinsically motivated, individuals curiously engage with their environments and reflect an inherent inclination toward learning and mastery (Ryan et al., 2021). Thus, in the new environment of online classes, students with higher levels of intrinsic motivation may also display higher levels of engagement (Hsu et al., 2019; Sun et al., 2019).

Furthermore, an important construct when navigating the new territory of online classes is self-efficacy, which captures the perceptions of competence (Guay et al., 2003). Specifically, self-efficacy has been defined as individuals' confidence level in their ability to organize and carry out a given course of action to accomplish tasks or solve problems (Bandura, 1986). Perceptions of self-efficacy are necessary for optimal functioning (Ryan & Deci, 2017) and have positive effects on online education (Prior et al., 2016). Thus, self-efficacy is an important part of successful online learning and even have a moderating effect on academic performance (De Feyter et al., 2012). Taken together, both intrinsic motivation and self-efficacy are important factors influencing students' engagement in online classes.

Present Study

The present studies consisted of two-wave longitudinal designs and followed two separate cohorts of university students over the Fall 2020 and Winter 2021 semesters to examine the impact of openness to experience and conscientiousness on engagement in online classes as well as the underlying motivational processes. There are countless ways for teachers to deliver online classes (Zhao, 2020; Zhao & Watterston, 2021). As teachers explore and gain experience with the novel online environment, they develop and adapt their classes accordingly. In this study, classes were predominantly conducted asynchronously (e.g., students can learn at their own pace) in contrast to synchronously (e.g., students receive immediate feedback). Both semesters were full terms, from early September to late December and from early January to late April, respectively. The Big 5 personality traits of openness to experience and conscientiousness, in addition to self-efficacy and motivation for online classes, were measured at the beginning of each semester, whereas engagement in online classes was measured at the end of each semester. The Fall and Winter samples were completely independent of each other.

The structuring of the current research was based on a recent study showing that openness to experience and conscientiousness were associated with heightened self-efficacy and autonomous motivation for online classes during the COVID-19 crisis (Audet et al., 2021). However, the present manuscript included an additional data set (Winter 2021) that has been collected since the Audet et al. (2021) paper was submitted. An expanded collection of data is advantageous since it provided the opportunity to assess changes over time. To the best of our knowledge, this is the first article to specifically focus on the role of openness to experience and conscientiousness on engagement with online classes during the entire remote academic year.

Based on the aforementioned literature, we first hypothesized that the relations between openness to experience and engagement in online classes would depend on the semester. We expected openness to experience to play

a major role in the Fall semester when online classes were novel and required adaptation. However, openness to experience may have minimal impact on engagement in online classes during the following Winter semester since the online environment would no longer be considered novel. Second, we expected conscientiousness to positively predict university students' engagement in online classes both in the Fall and Winter semesters. Finally, following SDT research, exploratory analyses were conducted to test whether motivational processes, such as the levels of intrinsic motivation and self-efficacy, would aid in explaining the associations between the two personality traits and engagement in online classes.

Method

Participants and Procedure

Participants were 350 students (M_{age} = 19.75, SD_{age} = 1.40; 87.8% female) for the Fall semester and 323 students (M_{age} = 19.45, SD_{age} = 1.53; 84.2% female) for the Winter semester at a large Canadian university.[1] All participants were recruited through a voluntary extra-credit subject pool. Participants completed surveys through a secure online survey software (Qualtrics) either in September and December (Fall 2020; overall retention rate of 48%) or in January and April (Winter 2021; overall retention rate of 43%). In the Fall semester, the ethnic/cultural background of our sample was predominantly European descent (56%) and included 29% of Asian descent, 8% of Middle Eastern descent, 2% of Hispanic descent, 1% of African descent, and 4% had another ethnic/cultural background. 61% of the participants were living with family during the semester, 25% with friends, and 14% alone.[2] In the Winter semester, the ethnic/cultural background of our sample was predominantly European descent (60%) and included 25% of Asian descent, 6% of Middle Eastern descent, 3% of Hispanic descent, 2% of African descent, and 4% had another ethnic/cultural background. 55% of the participants were living with family during the semester, 30% with friends, 10% alone, and 5% declined to answer.[3] The groups did not differ in age or self-reported gender. The study was approved by the University Research and Ethics Board.

Measures

Big 5 Personality Traits

At baseline (September or January), participants completed the 44-item Big five inventory (BFI) to evaluate their position on the Big 5 personality traits of conscientiousness, extraversion, neuroticism, agreeableness, and openness to experience (John & Srivastava, 1999). As presented above, the personality traits of openness to experience (measured with 9 items) and

conscientiousness (measured with 10 items) were the focus of the present study. Each item was rated based on how much participants felt it reflected their own personality on a scale from 1 (*strongly disagree*) to 5 (*strongly agree*). For example, an item used to assess openness to experience was "Is curious about many different things" and an item used to assess conscientiousness was "Does things efficiently." The reliability for the Big 5 personality traits was adequate, both during the Fall and Winter semesters, α's >.80. Combined, the reliability was α = .80 for openness to experience and α = .84 for conscientiousness.

Intrinsic Motivation for Online Classes

At baseline (September or January), students' intrinsic motivation for engagement in online classes was assessed using a scale adapted from Sheldon and Kasser (1998). Participants were asked to rate the extent to which they tried their hardest in their online classes for intrinsic reasons, with the item "Because of the fun and enjoyment which it provides you—the primary reason is simply your interest in the experience itself." This item was rated on a seven-point scale ranging from 1 (*strongly disagree*) to 7 (*strongly agree*).

Self-Efficacy for Online Classes

At baseline (September or January), students' self-efficacy for engagement in online classes was rated with the item "To what extent do you feel you have the skills and resources necessary to manage online education?" This item was rated on a seven-point scale ranging from 1 (*strongly disagree*) to 7 (*strongly agree*). This item was written for the purpose of the study and based on Shen et al.'s (2013) study of self-efficacy in online learning environments.

Engagement in Online Classes

Engagement in online classes was assessed at the end of each term (December or April) with 2 items: "Did you enjoy online learning?" and "To what extent would you rate your level of engagement with course materials (i.e., listening/assisting to lectures, doing readings, etc.)?" The same seven-point scale was used to assess both items, each using a seven-point scale ranging from 1 (*not at all*) to 7 (*very much*). These items were written for the purpose of the study based on discussions with a research team of undergraduate students. The reliability of the scale was α = .59 in the Fall semester and α = .63 in the Winter semester.

Analytic Strategy

Correlations and hierarchical regressions were conducted with SPSS statistics software (Version 28). Preliminary correlational analyses were conducted to show the means and correlations for the key variables in the study (openness

to experience, conscientiousness, intrinsic motivation, self-efficacy, and engagement in online classes). Independent sample *t*-tests were also conducted to compare the key variables in the two samples. The central results present three hierarchical multiple regressions in which self-efficacy, intrinsic motivation, and engagement in online classes were regressed on the semester (entered first), openness to experience and conscientiousness (entered second), and the interactions of semesters with both personality traits (entered third).

Furthermore, mediation analyses were performed on the associations that proved to be significant in the regressions. We used the method outlined by Hayes (2018) to test these mediation models by estimating 95% confidence intervals (CI) of the indirect effect using bootstrap resampling (k = 5,000) procedures.

Results

Preliminary Results

Data screening found the variables of interest to be normally distributed, making the variables suitable for regression analyses. Preliminary analyses revealed no effects approaching significance, p's > .20, for self-reported gender, ethnicity/cultural background, living circumstances, or age of participants on any of the dependent variables, thus we did not include these demographic factors in the main analyses.

Table 1 presents the means and standard deviations for self-efficacy, intrinsic motivation, and engagement in online classes for both the Fall and Winter semesters. Self-efficacy was relatively high, whereas intrinsic motivation and engagement in online classes were moderate. Comparisons between the Fall and Winter semesters suggest that self-efficacy was significantly higher in the Winter semester than in the Fall semester, t (676) = 1.98, p = .048, whereas intrinsic motivation was significantly higher in the Fall semester than the Winter semester, t (676) = 3.77, p < .001. Engagement in online classes did not differ across the two semesters, t (321) = −1.35, p = .180.

Table 1. Means and standard deviations among all measures by semester.

	n	M	SD
Self-efficacy			
1. Fall	346	5.34	1.38
2. Winter	332	5.55	1.31
Intrinsic motivation			
1. Fall	345	4.32	1.84
2. Winter	333	3.80	1.74
Engagement in online classes			
1. Fall	164	4.10	1.46
2. Winter	159	4.35	1.88

Table 2. Correlations among variables and residual change on outcomes by semester.

Variables	1.	2.	3.	4.	5.
1. Conscientiousness	-				
2. Openness to experience	.01	-			
3. Self-efficacy	.27**	.08	-		
4. Intrinsic motivation	.16**	.19**	.25*	-	
5. Engagement in online classes	.14*	.15**	.33**	.19**	-

** $p < .01$; * $p < .05$.

Next, correlational analyses suggest that self-efficacy and intrinsic motivation were significantly related and that both were significantly related to engagement in online classes at the end of semesters. The correlations also suggest associations between (1) openness to experience and intrinsic motivation and (2) conscientiousness and self-efficacy. Table 2 shows the correlations among variables and residual changes in outcomes by semester.

Main Results

Engagement in Online Classes

The regression of engagement in online classes suggest a significant multiple R of .30, F (5, 322) = 6.11, p < 001. No main effect was obtained for semester, $b = -.08$, t (325) = 1.42, $p = .643$. Significant effects emerged for openness to experience, $b = .19$, t (674) = 5.06, $p < .001$ and for conscientiousness, $b = .14$, t (322) = 2.51, $p = .013$. Semester moderated the relation of both openness to experience and conscientiousness with engagement in online classes but in opposite ways: openness to experience, $b = -.12$, t (322) = −2.13, $p = .033$, and conscientiousness, $b = .17$, t (322) = 3.18, $p = .002$. Subgroup analyses suggest that openness to experience was significantly related to engagement in online classes in the Fall semester, b =.29, t (161) = 3.78, $p < .001$, but not in the Winter semester, $b = .03$, t (161) = 0.70, $p = .713$. By contrast, conscientiousness was unrelated to engagement in online classes in the Fall semester, $b = -.05$, t (161) = −0.72, $p = .476$, but was significantly related in the Winter semester, $b = .28$, t (156) = 3.66, $p < .001$.

Self-Efficacy

The regression of self-efficacy suggests a significant multiple R of .30, F (5, 672) = 13.84, p < 001. A significant main effect emerged for semester, $b = .10$, t (675) = 2.56, $p < .011$. In addition, a significant main effect did not emerge for openness to experience, $b = .07$, t (674) = 1.87, $p = .061$, but did for conscientiousness, $b = .27$, t (674) = 7.38, $p < .001$. Whereas a significant interaction effect emerged for openness to experience with semester, $b = -.09$, t (674) = −2.46, $p = .014$, it did not for conscientiousness to self-efficacy, $b = -.01$, t (674) = −0.38, $p = .735$. Subgroup analyses suggest that openness to experience was significantly related to self-efficacy in the Fall semester, $b = .16$,

t (343) = 3.02, p = .003, but not in the Winter semester, b = −.02, t (328) = −0.33, p = .740.

Intrinsic Motivation

The regression of intrinsic motivation on engagement in online classes revealed a significant multiple R of .29, F (5, 672) = 12.51, p < 001. A significant main effect emerged for semester, b = −.14, t (675) = −3.69, p < .001. Additionally, a significant main effect emerged for openness to experience, b = .19, t (674) = 5.06, p < .001 and for conscientiousness, b = .15, t (674) = 4.05, p < .001. A significant interaction effect emerged for openness to experience with semester, b = −.07, t (674) = −1.98, p = .048, but not for conscientiousness to intrinsic motivation, b = .03, t (674) = 0.71, p = .481. Subgroup analyses suggest that openness to experience may be more strongly related to intrinsic motivation in the Fall semester, b = .25, t (343) = 5.82, p = .001, than in the Winter semester, b = .13, t (328) = 2.38, p = .017.

Mediation Results

Next, mediation analyses were performed to test whether the relation of openness to experience to higher engagement in online classes was mediated by intrinsic motivation, and whether the relation of conscientiousness to higher engagement in online classes may be mediated by self-efficacy. The measures of the Fall 2020 and Winter 2021 semesters were combined to simplify the presentation.

Results from the first mediation analyses (Figure 1) suggest that openness to experience was significantly related to intrinsic motivation, t = 2.90, p = .004. Intrinsic motivation was significantly related to engagement in online classes at the end of the semester, t = 3.12, p = .002. The total effect of openness to experience on engagement in online classes was significant at t = 2.78, p = .006. The indirect effect of openness to experience on engagement in online classes through intrinsic motivation was estimated to be b = .05, SE = .02, 95% CI [.01, .11]. This is considered significant since

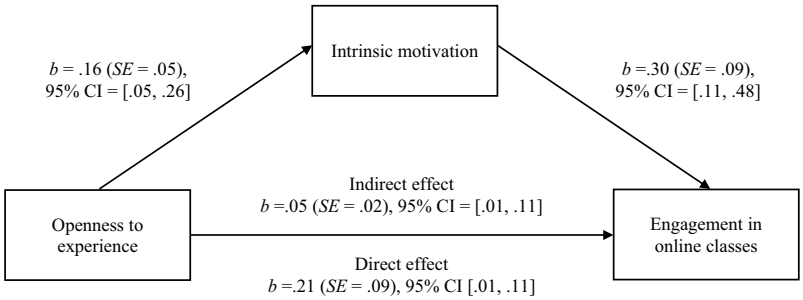

Figure 1. Direct and indirect effects of openness on participants' engagement in online classes.

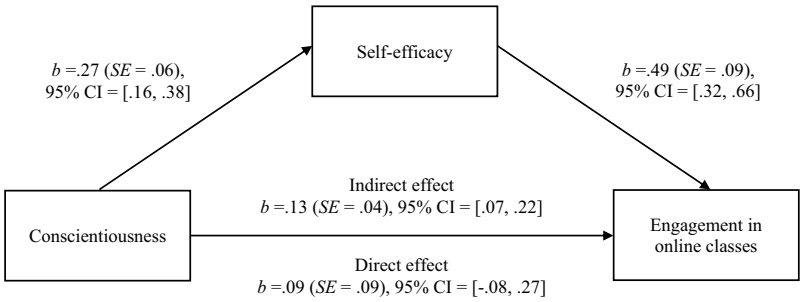

Figure 2. Direct and indirect effects of conscientiousness on participants' engagement in online classes.

the confidence intervals do not straddle zero (Hayes, 2018). The direct effects of openness to experience were $t = 2.28$, $p = .023$, suggesting partial mediation.

A second mediation analysis was performed to examine whether self-efficacy mediated the association between conscientiousness and engagement in online classes. Results from the mediation analyses (Figure 2) suggest that conscientiousness was significantly related with self-efficacy, $t = 4.76$, $p < .001$. Furthermore, self-efficacy was significantly related to engagement in online classes, $t = 5.70$, $p < .001$. The total effect of conscientiousness on engagement in online classes was significant at $b = .23$, SE = .09, $t = 2.48$, $p = .014$, 95% CI [.05, .41]. The indirect effect of conscientiousness on engagement in online classes through self-efficacy was estimated to be $b = .13$, SE = .04, 95% CI [.07, .22]. This is considered significant since the confidence intervals do not straddle zero (Hayes, 2018). The direct effect of conscientiousness was no longer significant, $t = 1.05$, $p = .296$, suggesting full mediation.

Discussion

The COVID-19 crisis transformed how university classes were taught during the 2020–2021 academic year. The present study used two prospective longitudinal samples to examine the predictive role of openness to experience (defined as intellectual curiosity, the tendency to seek novel experiences, the ability to adjust to change) and conscientiousness (defined as the tendency to be organized, reliable, and responsible) on university students' engagement in online classes. Students were also asked to report on their levels of intrinsic motivation (defined as enacting a behavior for the pure interest, enjoyment, and inherent satisfaction of the activity itself) and self-efficacy (defined as confidence in the ability to organize, carry out, and accomplish tasks) since these have positive effects on online learning (Hsu et al., 2019; Prior et al., 2016).

Openness to Experience and Consciousness

Our results suggest that openness to experience played a pivotal role in allowing university students to successfully engage in online classes during the COVID-19 crisis. As such, students who were higher in openness to experience reported higher levels of engagement in online classes at the end of the first online semester (Fall 2020). However, the advantage of openness to experience appeared to wash away during the subsequent Winter semester when online classes were no longer novel and had become more ordinary, suggesting that the Fall semester may have served as a "training" period for the students (Barrick & Mount, 1991).

The results also showed that the usual positive association between conscientiousness and academic outcomes was temporarily mitigated in the Fall semester. It was only during the second online semester (Winter 2021), and not the first, that conscientiousness was significantly positively related to engagement in online classes. This finding is surprising and not redundant with earlier research. This may be explained by the unique circumstance of a global health crisis in which the study was conducted, which left students no choice but to quickly transition into an online academic environment that was unknown and destabilizing. For instance, following the transition to online education in March 2020, students reported facing various barriers beyond their control to participation in online classes, such as COVID-19 related worries, internet problems, or unfavorable living situations, that could have persisted in the Fall semester (Gillis & Krull, 2020). Furthermore, it seems that the personality trait of conscientiousness shows its most potent effects in orderly and well-structured settings that have clear social expectations (Wilmot & Ones, 2019).

Motivational Processes

By asking students to report on their levels of intrinsic motivation and self-efficacy regarding engagement in online classes, we were able to gain insight into the motivational processes that may have accounted for the divergent results for openness to experience and conscientiousness. Process analyses confirmed that the association of openness to experience with engagement in online classes was partially mediated by intrinsic motivation (Holding & Koestner, 2023), whereas the association of conscientiousness with engagement in online classes was fully mediated by self-efficacy (Bandura, 1986). We did not expect such a particular association of openness to experience with intrinsic motivation and conscientiousness with self-efficacy. These motivational factors were included because a wealth of research has confirmed their importance in predicting school adjustment (Alivernini & Lucidi, 2011; Oriol-

Granado et al., 2017). Still, these interesting results perhaps could have been anticipated based on previous research (e.g., Şahin & Çetin, 2017; Sun et al., 2019).

Intrinsic Motivation

Indeed, our findings suggest a positive relation between intrinsic motivation and engagement with online classes, which supports prior research (Sun et al., 2019). Similarly, the specific pairing of openness to experience with intrinsic motivation is not new. For example, openness to experience has been found to predict intrinsic motivation (Hazrati-Viari et al., 2012) and studies done with students (Du et al., 2019; Tan et al., 2019) and employees (Watanabe & Kanazawa, 2009) found that high levels of openness to experience were related to high levels of intrinsic motivation. The interesting association between the two may be partially explained by curiosity (Tan et al., 2019). Indeed, openness to experience is characterized as an intellectual curiosity (Schmutte & Ryff, 1997) and seeking to explore new things (McAdams, 2015), whereas intrinsic motivation is characterized as curiously engaging in ones' environment (Ryan et al., 2021). Thus, students higher in openness to experience seemed to have an inherent curiosity to engage with the new academic setting and perhaps took on a greater interest in the unfamiliar methods of online classes.

Self-Efficacy

Second, results suggest a positive relation between self-efficacy and engagement with online classes. This supports prior studies depicting that self-efficacy is an important component of being successful in online education environments (Prior et al., 2016). In addition, the relation between conscientiousness and self-efficacy could have been expected as several previous studies have found links between the two (Chen et al., 2020; De Feyter et al., 2012; Şahin & Çetin, 2017). Interestingly, our findings suggest that the relation only emerged in the second online semester. This may be explained by the lack of predictability and structure that accompanied the COVID-19 crisis. Since self-efficacy refers to individuals' confidence in their ability to carry out a task (Bandura, 1986), students higher in conscientiousness may have had to experience some small victories with online classes (e.g., succeeding at online exams, presentations, or essays) before gaining confidence in their abilities to flourish in an online environment. Taken together, the specific relations of openness to experience with intrinsic motivation and conscientiousness with self-efficacy are in line with previous literature.

Limitations

Our results should be considered alongside their limitations. First, our sample size was small, predominantly female and of European or Asian descent, and was narrowly drawn from students at a highly competitive university during the COVID-19 crisis. The generalizability of our results thus remains unclear. Second, it is possible that the correlations in the present study were due to confounding variables (e.g., unanticipated technology problems and facing academic barriers related to the COVID-19 crisis, with students from marginalized backgrounds disproportionately experiencing these challenges) and we cannot rule out alternative explanations and clearly establish causality. Third, the data collected was based on self-reports. Nevertheless, self-reports do hold value and have been shown to yield similar results to peer reports and objective measures (Levine et al., 2019). Fourth, this study focuses exclusively on engagement in online classes and would be strengthened by adding academic achievements. Fifth, teachers' self-efficacy is not considered. However, teachers face numerous challenges when they transition to online environments (Horvitz et al., 2015). This task has been even more demanding for them due to the sudden implementation of online education which resulted in numerous technological difficulties (Rabaglietti et al., 2021). Indeed, the novel teaching requirements and demands have impacted teachers' self-efficacy (e.g., instructional efficacy; Pressley et al., 2021), which may in turn have influenced students' engagement in online classes. Finally, the use of brief scales must be highlighted because they are limited in their ability to fully capture the complexities of the constructs they assess. If time constraints and participant burden allow for it, future research should include more intricate measures of these variables.

Implications and Future Directions

Though not the focus of the current research, the results can also enhance educational settings following the COVID-19 crisis. Indeed, the unprecedented changes brought forth have magnified educational challenges and barriers, with students reporting increasing levels of adverse mental health outcomes since the start of the COVID-19 crisis (Banerjee & Rai, 2020; Czeiler et al., 2020; Hamza et al., 2021; Pietrabissa & Simpson, 2020).

Thus, educational service providers, such as health and counseling services, academic success management, and general services in universities, would gain from implementing the findings in academic services and classrooms across campuses. When developing new programs and offering services to students, it would be beneficial to promote educational environments favoring motivational factors based on the uniqueness of students. Educational environments that increase feelings of responsibility (e.g.,

focusing on tasks, organizational skills, goal setting and planning, etc.) would benefit students higher in conscientiousness, and educational environments that increase interest and curiosity (e.g., maintaining engagement, offering the possibility of making meaningful choices, etc.) would benefit those higher in openness to experiences. Such environments would increase students' sense of purpose and persistence in their studies. In addition, it would best suit and assist students during the numerous challenges they are encountering.

The findings highlight the fundamental importance of educational environments that foster intrinsic motivation and self-efficacy. Witnessing that the COVID-19 crisis is having long-lasting impacts on students (Banerjee & Rai, 2020; Hamza et al., 2021; Pietrabissa & Simpson, 2020), strengthening educational environments with these findings would benefit our current and future generations of students.

Conclusion

The dramatic alteration in how university classes were taught during the 2020–2021 academic year provided a unique opportunity to examine how the Big 5 personality traits of openness to experience and conscientiousness influenced students' engagement in online classes, both initially and over time. Our results suggest that during the first online semester, the personality trait of openness to experience allowed students to be interested and excited about the opportunity to study online, which resulted in heightened engagement in online classes. By contrast, once students adapted to the novel online environment, the results of the second online semester suggest that openness to experience was no longer associated with intrinsic motivation and heightened engagement in online classes. Instead, conscientiousness emerged as the key to heightened engagement in online classes. Finally, there was evidence of different mediational pathways for the two personality traits: openness to experience was related to intrinsic motivation whereas conscientiousness was related to self-efficacy.

Notes

1. The Fall 2020 and Winter 2021 datasets were completely independent of each other.
2. Living circumstances were unrelated to participants' scores on openness to experience, $p = .605$, conscientiousness, $p = .563$, intrinsic motivation, $p = .210$, and self-efficacy, $p = .402$.
3. Living circumstances were unrelated to participants' scores on openness to experience, $p = .640$, conscientiousness, $p = .627$, intrinsic motivation, $p = .259$, and self-efficacy, $p = .361$.

Disclosure Statement

No potential conflict of interest was reported by the author(s).

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: The preparation of this manuscript was supported by a Social Sciences and Humanities Research Council of Canada grant (1008-2020-0146).

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Élodie Audet is in her second year of graduate studies in the Clinical Psychology program at McGill University in the McGill Human Motivation Lab under the supervision of Dr. Richard Koestner. Élodie is currently pursuing research examining the way our social environments can enhance or undermine capacities for both regulation and integration of emotional experiences, and how this relates to individuals' motivation, personal goal pursuits, and well-being.

Shelby Levine is currently in her fifth year of graduate studies in the Clinical Psychology program at McGill University. She completed her M.A. in experimental psychology at Carleton University in April 2018 with Dr. Milyavskaya. Shelby studied the influence of perfectionism on mental health in the transition to university. Shelby also completed her undergraduate studies at Carleton University. Shelby's research program explores the collaborative nature of autonomy. How autonomy flourishes through collaboration in goal striving and how more collaborative individuals are able to gain support during goal pursuit and during crises.

Pascale Dubois is a speech-language pathologist who has worked for 12 years with adolescents and young adults with developmental language disorder and learning disabilities. In her research, she is interested in the school-to-work transition of students with disabilities and the role of the social environment during the transition to adulthood. She is currently a post-doctoral fellow working in the Human Motivation Lab.

Sophie Koestner is in her second year of the Bachelors of Social Work program at McGill University. Sophie is planning to do her stage in community development and hopes to work with senior populations in her future career.

Richard Koestner is a professor of Psychology at McGill University. He has dedicated more than 30 years of research to studying personality and human motivation. He is the founder and director of the McGill Human Motivation Lab, and he has published over 180 scientific articles. Dr. Koestner recently won the Canadian Psychology Association's Donald O. Hebb Award for Distinguished Contributions to Psychology as a Science (2021). Dr. Koestner is one of the world's leading proponents of Self-Determination theory and his recent work focuses on how to effectively set and pursue personal goals.

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Ethical Statement

The study was conducted in compliance with APA ethical standards.

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